A Robust Attitude and Heading Reference System Algorithm for MEMS-Based inertial sensors aided by Magnetometer using EKF

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Design of sophisticated Algorithms for MEMS Based Attitude and Heading Reference Systems (AHRS) is a hot topic in the research field of Navigation. MEMS based inertial sensors have great advantage of reducing entire cost of AHRS but low accuracy of such sensors need to be complemented with more intelligent algorithms to achieve reliable performance. In this dissertation, a Kalman Filter based, MATLAB algorithm for AHRS is developed. Gravity vector information from accelerometers and Earth's Magnetic field information measured by magnetometers is used for AHRS Aiding to improve attitude accuracy. Detailed analysis of AHRS performance has been done, in static and accelerated conditions, over simulated data. Finally the effectiveness of algorithm is tested over the real van test data.

Keywords: AHRS, gyroscopes, accelerometers, magnetometers, Kalman Filter

1. Introduction

The Inertial Measurement Units (IMUs) based on the MEMS technology includes the MEMS gyroscopes, accelerometers and MEMS magnetometers which have great applications in the field of navigation because of having small size and low cost. Attitude and Heading Reference System (AHRS) contain MEMS based IMU which have 9-degree of freedom containing triads of gyroscopes, accelerometers and magnetometers. If initial angles information is given then gyroscope measurements give us the orientation information in term of the heading (yaw), pitch and roll. But due to noise and bias of the low cost MEMS inertial sensors, errors grow rapidly in attitude angles and it drifts quickly from the actual orientation [3]. Another way of obtaining the attitude angles is by using the accelerometers and magnetometers together but there results also deviates for the dynamic system. T attain optimal results for AHRS best option is to integrate both the system i.e. gyros system with the accelerometers and magnetometers system. The gyros drift is estimated by the second system and after compensating, long term results can be achieved.

In AHRS algorithms, the Kalman filter is extensively used for the integration and gyro biases estimation. In order to use Kalman filter for AHRS one should have to develop the dynamic/kinematic and measurement model. In 1993 the different attitude representation are given by the Shuster which include Euler angle and quaternions [4]. Keeping in view these attitude representations the different Kalman filter based dynamic and the measurement models for AHRS were developed and the computations for the attitude i.e. orientation measurement have been done. It has been observed that the Euler angle based AHRS model's calculations have more accuracy than the quaternions based AHRS model's calculations [5]. But the quaternions based model have the most significant advantage that it has no singularity while the Euler angles based model have the singularity when the pitch passed through $\pm\pi/2$ [6].

In 1994 Euler angles based model was presented by the Emura and Tachi, they used

the Kalman filter based approach for attitude estimation. In that model the state vector was consists of six elements, three angular rates and three Euler angles components. As dynamic model was Euler angle based again the disadvantages of this method was that it has the singularity as pitch passes through $\pm\pi/2$ it diverge which is not affordable and also Euler angle kinematics are nonlinear derivative equations [7].

In 1996 Euler angle error based method was presented by Foxlin in which the Kalman filter was implemented. The state vector contain the six elements, three gyro biases and three Euler angle errors [8]. The same kind of model was again proposed by the Setoodeh et al in 2004. In this model the errors in body frame was used as Euler angle errors. Since the Euler angle method was again used so singularity is there and also nonlinearity [9]. To overcome this singularity in the Euler angle kinematic model should have to use the quaternions based method. Quaternions based kinematic model have no singularity in it and also the linear kinematic equations [10]. But the drawback of this method is that the measurement model of this is still nonlinear [11]. To overcome this nonlinearity the Extended Kalman filter (EKF) was taken as standard tool. The EKF takes more time for calculation of attitude than the usual KF because of nonlinearity errors. Another way to reduce the computational time is to pre-process the nonlinearities of the sensors and then using the simple direct KF as in 2002 Qi and Moore proposed this method. The drawback of this method is that performance parameters are affected strongly [12].

In 2011 a novel method to integrate IMU with magnetometers for AHRS was proposed by Han and Wang to overcome the nonlinearity problems. Although this model was a linear model based on the Euler angle errors but observation model suffer from singularity problem as the pitch passes through $\pm\pi/2$ [13]. In this thesis AHRS model is developed using EKF. The dynamics model is based on Euler angle errors which contain three Euler angle errors and three gyros biases in state vector. In all above discussed models no one have include the effect of sudden dynamics, acceleration step on the orientation parameters in this paper this effect is also included. The MATLAB AHRS algorithm is developed and tested over the simulated data in static condition and real test data.

2. Attitude determination using AHRS IMU

In this section methods to use the data coming from the AHRS IMU is presented to determine the orientation angles. There are two approaches of obtaining attitude angles i.e. first is relative approach of calculating attitude angles from the quaternions updating (using gyros angular rates) and second is absolute approach of calculating attitude angles (pitch and roll) from accelerometers and heading (yaw) from the magnetometers. But to initialize the quaternions the orientation angles are required which are obtained from accelerometers and magnetometers averaging. After initializing quaternions, the quaternions updating is started using gyros angular rates and the orientation angles from accelerometers and magnetometers, therefore let discuss it step by step.

2.1. Pitch and roll from Accelerometers

The data coming from the accelerometers is consist of the acceleration of body frame in all

three axis x, y and z-axis with respect to inertial frame also referred to as gravity vector in the body-frame [2] as follows

$$g^b = \underline{a} = [a_x, a_y, a_z]^T \quad (1)$$

The gravity vector in the navigation frame is given by

$$g^n = \begin{bmatrix} 0 & 0 & -g \end{bmatrix}^T \quad (2)$$

Where g is normally ~9.8m/s^2, the relationship between the gravity vector in the body frame and gravity frame in the navigation frame is given by using the DCM (which transforms the coordinates form n-frame to b-frame) as follows

$$g^b = C_n^b g^n \quad (3)$$

using equation (1), (2) in (3)

$$\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} cos\theta cos\psi & cos\theta sin\psi & -sin\theta \\ sin\varphi sin\theta cos\psi - cos\varphi sin\psi & cos\varphi cos\psi + sin\varphi sin\theta sin\psi & cos\theta sin\varphi \\ cos\varphi sin\theta cos\psi + sin\varphi sin\psi & -sin\varphi cos\psi + cos\varphi sin\theta sin\psi & cos\theta cos\varphi \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}$$

$$\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} -sin\theta \\ cos\theta sin\varphi \\ cos\theta cos\varphi \end{bmatrix} g \quad (4)$$

From equation (4) the pitch and roll can be calculated as

$$\begin{bmatrix} \varphi \\ \theta \end{bmatrix} = \begin{bmatrix} a\tan 2(a_y, a_z) \\ a\tan 2(-a_x, \sqrt{a_y^2 + a_z^2}) \end{bmatrix} \quad (5)$$

In practice the acceleration output can be averaged to yield as estimate of the gravity vector in the body frame [2]. Hence the estimate of the body frame gravity is

$$\bar{g}^b = \begin{bmatrix} \bar{g}_1 & \bar{g}_2 & \bar{g}_3 \end{bmatrix}^T = \frac{1}{T}\int_0^T -y_a(\tau)d\tau \quad (6)$$

Where y_a is output from accelerometers. Therefore using equation (5), (6) and (2.9) the pitch and roll from accelerometer can be determined as follows

$$\begin{bmatrix} \varphi \\ \theta \end{bmatrix} = \begin{bmatrix} a\tan 2(\bar{g}_2, \bar{g}_3) \\ a\tan 2(-\bar{g}_1, \sqrt{\bar{g}_2^2 + \bar{g}_3^2}) \end{bmatrix} \quad (7)$$

2.2. Magnetic Heading from magnetometers

The magnetic heading (yaw angle) can be calculated from the relation of magnetometers magnetic field in the b-frame and n-frame. Such relation is determined again by the using the navigation to body DCM $C_n^b = C_\varphi C_\theta C_\psi$ [2]. Hence the relation between the Earth magnetic field vector in body frame m^b and in the navigation frame m^n is given as

$$m^b = [C_\varphi][C_\theta][C_\psi]m^n \quad (8)$$

Decomposing the field vector from b-frame to n-frame plane by only the reverse (negative) rotation of the roll and pitch one can find relation for magnetic heading

$$[C_\theta]^T [C_\varphi]^T m^b = [C_\psi]m^n$$

Using the values of the (C_φ, C_θ) and magnetic field vector in the n-frame as $m^n = [m_e \quad 0 \quad 0]^T$

$$\begin{bmatrix} \cos\theta & \sin\theta\sin\varphi & \sin\theta\cos\varphi \\ 0 & cos\varphi & -sin\varphi \\ -\sin\theta & \cos\theta sin\varphi & \cos\theta cos\varphi \end{bmatrix} m^b = \begin{bmatrix} cos\psi \\ -sin\psi \\ 0 \end{bmatrix} m_e \quad (8)$$

Same like as accelerometers output is averaged the magnetometer output is also averaged to estimate the magnetic field vector in b-frame.

$$\bar{m}^b = \begin{bmatrix} \bar{m}_1 & \bar{m}_2 & \bar{m}_3 \end{bmatrix}^T = \frac{1}{T}\int_0^T y_m(\tau)d\tau \quad (9)$$

y_m is the output from the magnetometers. Putting value magnetic field vector from equation (9) in (8)

$$\begin{bmatrix} \bar{m}_1^w \\ \bar{m}_2^w \\ \bar{m}_3^w \end{bmatrix} = \begin{bmatrix} cos\psi \\ -sin\psi \\ 0 \end{bmatrix} m_e \quad (10)$$

Hence from above equation (2.15) magnetic heading is given by

$$\tan\psi = \frac{-\bar{m}_2^w}{\bar{m}_1^w} \quad \gg \quad \psi = a\tan 2(-\bar{m}_2^w, \bar{m}_1^w) \quad (11)$$

The magnetic heading (yaw) from magnetometers, pitch and roll from

accelerometers are used to initialize the quaternions (q_0, q_1, q_2, q_3) [2].

3. Attitude from quaternions updating (from gyros)

After the initialization quaternions are updated repeatedly with respect to time and yaw pitch and roll are determined which are briefly discussed in this section. The angular rates of the navigation frame (n-frame) with respect to the body frame (b-frame) represented in the body frame [2] can be written as

$$\omega_{bn}^{b} = [\omega_1, \omega_2, \omega_3]^T = [\omega_x, \omega_y, \omega_z]^T \quad (11)$$

The quaternion's differential equation [1] can be written in the matrix form

$$\begin{cases} \dot{q}_0 = -\frac{1}{2}\left(q_1\omega_x + q_1\omega_y + q_1\omega_z\right) \\ \dot{q}_1 = \frac{1}{2}\left(q_0\omega_x + q_2\omega_z - q_3\omega_y\right) \\ \dot{q}_2 = \frac{1}{2}\left(q_0\omega_y - q_1\omega_z + q_3\omega_x\right) \\ \dot{q}_3 = \frac{1}{2}\left(q_0\omega_z + q_1\omega_y - q_2\omega_x\right) \end{cases} \quad (12)$$

In the matrix form

$$\dot{\mathbf{q}} = \frac{1}{2}\begin{bmatrix} -q2 & -q3 & -q4 \\ q1 & q4 & -q3 \\ -q4 & q1 & q2 \\ q3 & -q2 & -q1 \end{bmatrix}\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \quad (13)$$

Using the angular from the three gyros in IMU placed in each perpendicular axis i.e. $\omega_{bn}^{b} = [\omega_x, \omega_y, \omega_z]^T$ first the quaternions are updated at each time step using above set of equation (13) and then using these quaternions each time DCM is determined as quaternion to DCM conversion is given by

$$DCM = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2q_1q_2 - 2q_0q_3 & 2q_0q_2 - 2q_1q_3 \\ 2q_0q_3 + 2q_1q_2 & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2q_2q_3 - 2q_0q_1 \\ 2q_1q_3 - 2q_0q_2 & 2q_0q_1 + 2q_2q_3 & q_0^2 - q_1^2 - q_1^2 + q_3^2 \end{pmatrix} \quad (14)$$

Then the orientation angles are determined by comparing the above DCM from quaternion to the DCM from Euler angles which also previously have been derived. The Orientation angles are determined by following formulas

$$\begin{bmatrix} \psi \\ \theta \\ \varphi \end{bmatrix} = \begin{bmatrix} a\tan 2(2q_1q_2 - 2q_0q_3 \;,\; q_0^2 + q_1^2 - q_2^2 - q_3^2) \\ \sin^{-1}(-(2q_0q_2 + 2q_1q_3)) \\ a\tan 2(2q_2q_3 - 2q_0q_1 \;,\; q_0^2 - q_1^2 - q_2^2 + q_3^2) \end{bmatrix} \quad (15)$$

4. Kalman Filter Scheme Mechanization

4.1. State Vector

In our AHRS dynamic model the elements of the state vector are the errors in pitch, roll and the gyro biases as follows

$$\underline{x} = \left[\delta\psi \quad \delta\theta \quad \delta\varphi \quad \delta p \quad \delta q \quad \delta r\right]^T \quad (16)$$

Where $\delta\dot{\theta}$ and $\delta\dot{\varphi}$ are the uncertainties in the pitch and roll $\delta\dot{p}$, $\delta\dot{q}$ and $\delta\dot{r}$ are the uncertainties in the components of the angular velocity. As state vector have two values at the same time, that is the a priori value, the predicted value before the update, and the a posteriori value, the corrected value after the update. In the following a priori value is marked by $\underline{x}^{-}(t)$ and a posteriori value is marked by $\underline{x}^{+}(t)$.

4.2. Dynamics Model

To develop error dynamic model for AHRS, expression for the rates of attitude angles i.e. $\delta\dot{\psi}$ $\delta\dot{\theta}$ and $\delta\dot{\varphi}$ can be derived using equations (2.44), (2.45) and (2.46), which are

$$\dot{\psi} = (1/\cos\theta)q\sin\varphi + r\cos\varphi \quad (17)$$

$$\dot{\theta} = q\cos\varphi - r\sin\varphi \quad (18)$$

$$\dot{\varphi} = p + \tan\theta(q\sin\varphi + r\cos\varphi) \quad (19)$$

Taking the differential of above equations

$$\boxed{\begin{aligned}\delta\dot{\psi} &= (\sec\theta\tan\theta)(q\sin\varphi + r\cos\varphi)\delta\theta \\ &+ (\sec\theta)(q\cos\varphi - r\sin\varphi)\delta\varphi \\ &+ \sec\theta\sin\varphi\delta q + \sec\theta\cos\varphi\delta r\end{aligned}} \quad (20)$$

$$\boxed{\begin{aligned}\delta\dot{\theta} &= (-q\sin\varphi - r\cos\varphi)\delta\varphi \\ &+ (\cos\varphi)\delta q - (\sin\varphi)\delta r\end{aligned}} \quad (21)$$

$$\boxed{\begin{aligned}\delta\dot{\varphi} &= (q\sin\varphi + r\cos\varphi)\sec^2\theta\delta\theta \\ &+ (q\cos\varphi - r\sin\varphi)\tan\theta\delta\varphi + \delta p \\ &+ \sin\varphi\tan\theta\delta q + \cos\varphi\tan\theta\delta r\end{aligned}} \quad (22)$$

The gyro biases $\delta\dot{p}$, $\delta\dot{q}$ and $\delta\dot{r}$ are modelled as

$$\left.\begin{aligned}\delta\dot{p} &= -\beta\delta p + w \\ \delta\dot{q} &= -\beta\delta q + w \\ \delta\dot{r} &= -\beta\delta r + w\end{aligned}\right\} \quad (22)$$

$\beta = 1/\tau$ is inverse of correlation time. The general equation for the dynamic model is given as

$$\dot{\underline{x}}(t_i) = F\,\underline{x}(t_{i-1}) + w \quad (23)$$

$$\delta\dot{\underline{x}}(t_i) = F\,\delta\underline{x}(t_{i-1}) + w \quad (24)$$

The F matrix is given as

$$F = \begin{bmatrix} 0 & (\sec\theta\tan\theta)(q\sin\varphi + r\cos\varphi) & (\sec\theta)(q\cos\varphi - r\sin\varphi) & 0 & \sec\theta\sin\varphi & \sec\theta\cos\varphi \\ 0 & 0 & (-q\sin\varphi - r\cos\varphi) & 0 & \cos\varphi & -\sin\varphi \\ 0 & \sec^2\theta(q\sin\varphi + r\cos\varphi) & (q\cos\varphi - r\sin\varphi) & 1 & \sin\varphi\tan\theta & \cos\varphi\tan\theta \\ 0 & 0 & 0 & -\beta & 0 & 0 \\ 0 & 0 & 0 & 0 & -\beta & 0 \\ 0 & 0 & 0 & 0 & 0 & -\beta \end{bmatrix} dt$$

4.3. Covariance matrix for dynamic model

The variance of the variables of state vectors is govern by the covariance matrix of the dynamic model known as dynamic process noise covariance matrix usually denoted by Q. For this AHRS dynamic model the general Q is a 6x6 matrix having the variance of each of the variable of state vector (variance of heading, pitch, roll and three gyros biases) at the diagonals and the off diagonal elements are zero as follows.

$$Q = \begin{bmatrix} \sigma_\psi^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_\theta^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_\varphi^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_p^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_q^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma_r^2 \end{bmatrix} \quad (25)$$

Note that if the variances (σ_p σ_q σ_r) in the Q matrix have less value then its mean that gyroscopes have less standard deviation and high accuracy then more weightage is given to the gyroscopes output attitude angles in the Kalman filter algorithm.

4.4. Measurement (Observation) Model

The observation model represents the relationship between the state and the measurements. In the linear case the measurements can be described by a system of linear equations, which depend on the state variables.

$$\underline{z}_k = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta\psi \\ \delta\theta \\ \delta\varphi \\ \delta p \\ \delta q \\ \delta r \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ v_1 \\ v_2 \\ v_3 \end{bmatrix} \quad (26)$$

4.5. Covariance matrix for measurement model

The variance of the measurement variables is governed by the measurement noise covariance matrix usually denoted by R. For this AHRS model R is a 3x3 matrix having the standard deviation of measurement (heading pitch and roll) at the diagonal and off diagonal elements are zero as follows.

$$R = \begin{bmatrix} \sigma_{\psi}^2 & 0 & 0 \\ 0 & \sigma_{\theta}^2 & 0 \\ 0 & 0 & \sigma_{\varphi}^2 \end{bmatrix} \quad (27)$$

The value of R defines that how much weightage should be given to the measurement in the Kalman filter algorithm for AHRS. Larger the value of R means the measurements have large deviation so the less weightage would be given and vice versa. The measurement accuracy used in this paper for real time data of MEMS inertial sensors was 1 degree for heading and 0.5 degree for pitch and roll.

5. Implementation of Kalman filter

The flow chart is given in following figure from which it is clear that first of all attitude angles are computed from angular rates of gyroscopes using differential equations of quaternions (quaternions updating).

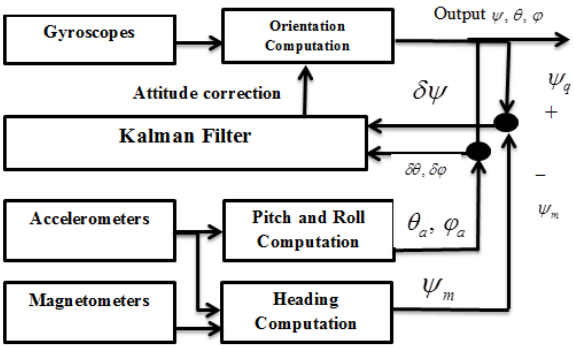

Figure 1: AHRS algorithm flow chart with Kalman filter

Then another pitch and roll are computed from accelerometers using gravity vectors discussed earlier, these values of pitch and roll are used for the computation of magnetic from magnetometers. The two types of attitude angles have been computed one from gyroscopes and second from accelerometers and magnetometers. The difference of these two angles is defined as measurements, which are provided to the Kalman filter, and after the best estimation of state, the attitude correction is done.

6. Results and Discussion

Detailed analysis of AHRS algorithm have been done which includes attitude angles & gyroscopes biases estimation, AHRS algorithm performance in accelerated environment, identification of system dynamics and setting the values of process noise covariance matrix Q and measurement noise covariance matrix R accordingly. Finally, the AHRS algorithm is tested out over the simulated flight test-data and real van test-data.

6.1. Attitude angles and Gyro bias estimation over simulated data in static condition

The gyroscope biases changes over time, it is necessary to take care of gyro biases for the practical implementation of every IMU which includes gyros. The gyro drift is the main source of error in an AHRS. As Kalman filter based approach is used in our AHRS algorithm, one advantage of this approach is that it can estimate the gyroscope biases with very good accuracy as an additional state in the system dynamic model.

Proof of Concept in static condition

The gyro bias can be estimated by using Kalman filter based AHRS algorithm in static condition of vehicle. First gyro biases are estimated in static condition by giving the known gyro biases as input to the AHRS algorithm and checking output whether it gives the right gyro biases or not, and also the known heading pitch and roll in static condition.

Case: 1

In the first case it has been assumed that the vehicle is in known static condition having known heading pitch and roll of 15°, 10° and 5°

respectively with respect to geographic frame which taken as navigation frame. In static condition the accelerometers reads normal reaction to gravity Acc = [0; 0; -9.8] in geographic frame and for sensor data simulation this acceleration vector is transformed into the body frame by the DCM obtained from the given static attitude angles. The magnetic field measured by the magnetometers is taken constant at the place where our vehicle is standing. Input gyro-biases in all three gyros are taken as *wx_p = 100 deg/h, wy_q = 200 deg/h, wz_r = 300 deg/h.* Now the sensor data simulation for static condition and necessary things to run the AHRS algorithm are completed. Using this data the attitude angles and the gyro biases are estimated as shown in the following figure (3.1). Remember these results are without incorporating the earth rotation rates in AHRS algorithm.

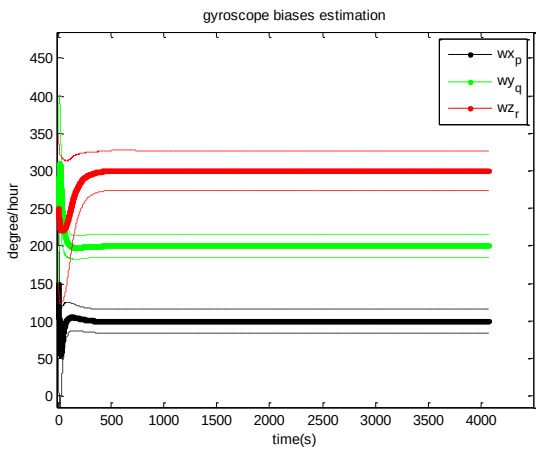

Figure 2: Gyroscope biases estimated along with variance band (accuracy)

Hence from above results gyro biases are estimated same as input was given so AHRS algorithm works properly in static condition. Remember that it takes some short time to match estimated biases with true known biases because accuracy is optimized within the limit of 25 deg/h offset by setting the value of Q matrix.

The initial heading, pitch and roll are [15°, 10°, and 5°] in static condition and estimated by AHRS algorithm are exactly matched as shown in the figure (3) & figure (4).

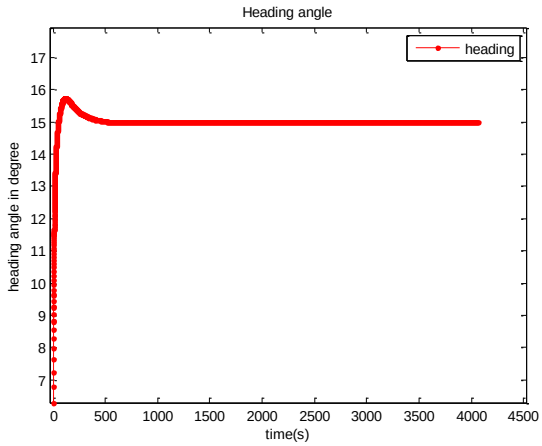

Figure 3: Estimated of heading angle in static condition

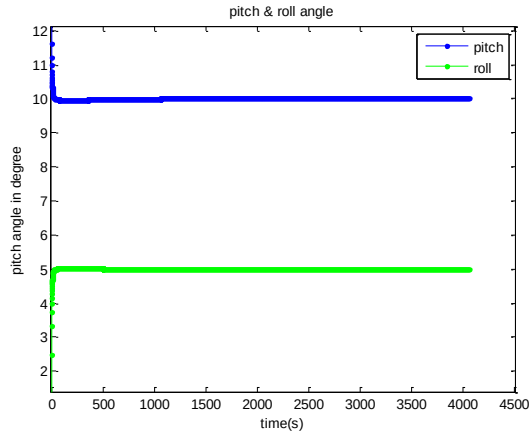

Figure4: Estimated pitch and roll in static condition

Case: 2 (introducing bias & noise in gyros and accelerometers)

In this case a Gaussian noise of 5mg is introduced in the simulated accelerometer data along with the bias of 15mg in addition to this,

Input biases in the gyros

wx_p = 100, wy_q = 200, wz_r = 300. (deg/h)

Noise in the gyros → 0.05 deg/s (1σ Gaussian)

Bias in accelerometers →15mg

Noise in accelerometers →5mg (1σ, Gaussian)

The sensor data is simulated which is almost close to the real IMU data. Now one can check out the results of our AHRS algorithm. The results for gyros biases are shown in the figure (5).

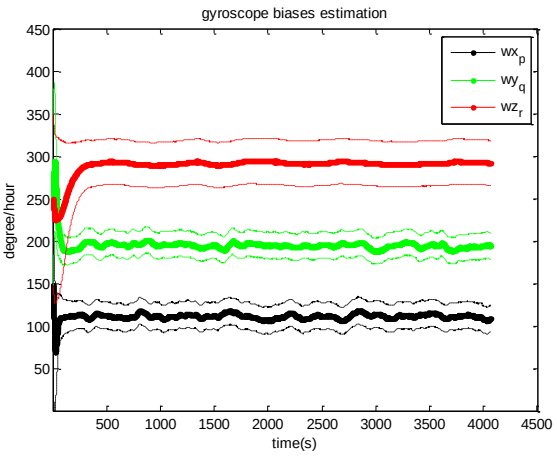

Figure 5: Estimation of gyro biases including noise in the accelerometers

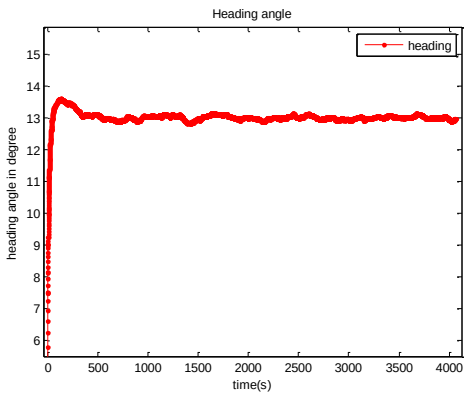

Figure 6: Estimated heading with bias + noise in gyros and accelerometers

By introducing noise in accelerometers, gyro biases are changed slightly but attitude angles are shifted 1-2 degree. Therefore one can get the results of attitude angle within deviation of 1-2 degree, noise up to 5mg is filtered by the Kalman filter.

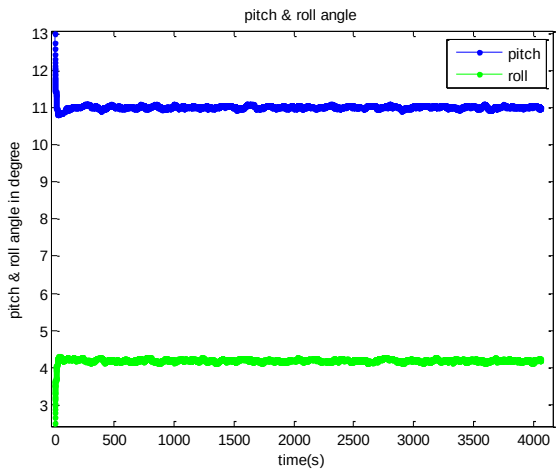

Figure 7: Estimated pitch and roll with bias + noise in gyros and accelerometers

6.2. Van test data analysis

In van test data analysis AHRS algorithm is tested out for estimation of the attitude angles of van as it moves. The tuning parameters of Kalman filter i.e. process noise covariance matrix Q and measurement noise covariance matrices R for van test are set out as

$$Q = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2_{gyro}(1-e^{-2\beta\Delta t}) & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma^2_{gyro}(1-e^{-2\beta\Delta t}) & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma^2_{gyro}(1-e^{-2\beta\Delta t}) \end{bmatrix}, \quad R = \begin{bmatrix} (1*d2r)^2 & 0 & 0 \\ 0 & (0.5*d2r)^2 & 0 \\ 0 & 0 & (0.5*d2r)^2 \end{bmatrix}$$

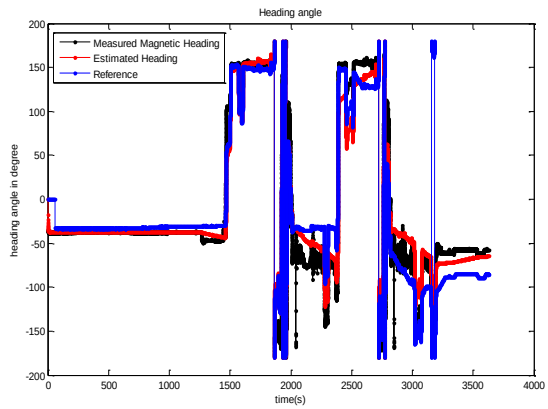

Figure 8: Estimation of heading angle for van test

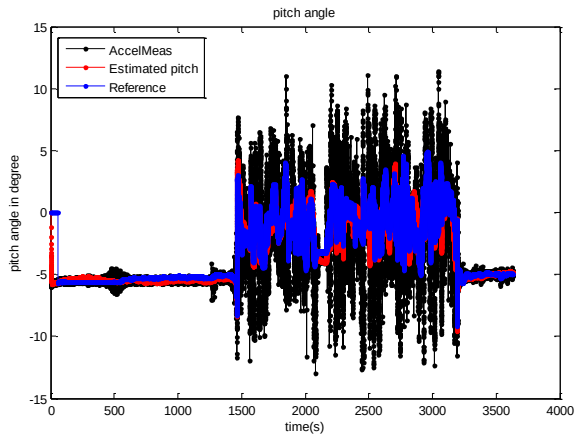

Figure 9: Estimation of pitch angle for van test

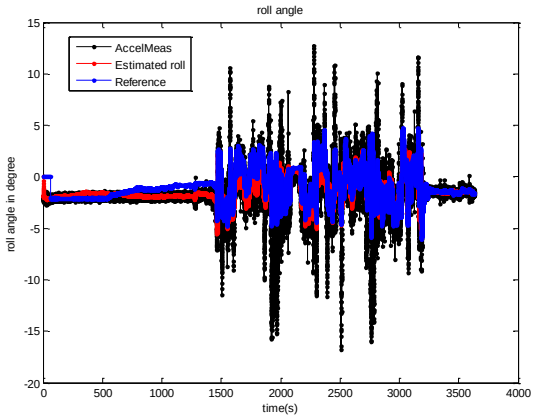

Figure10: Estimation of roll angle for van test

From above results of heading in figures (8) our estimation indicated by red line matches with the magnetic north indicated by black line but deviates somehow from reference true north indicated by blue line. This is because true north and magnetic north are not aligned exactly there is a declination angle between them. If declination angle is incorporated too, the results would be much better. From tuning point of view if the values in diagonal of Q matrix are of the order of 1e-6 then our estimation approaches slowly to magnetic heading, that is required in actual situation because the magnetic heading information are not more reliable due to disturbance of magnetic field of vehicle itself. The pitch and roll are estimated with very good accuracy and noise is removed as well as shown in above figures (9) & (10) respectively. In Future work this AHRS algorithm can be integrated with GPS/INS for the complete navigation.

7. Conclusion

The main objective was to develop a robust AHRS algorithm for MEMS-based inertial sensors, aided with magnetometers, to attain the optimal estimate of attitude angles in different dynamic environments. Based on Euler Angle formulation, linearized error dynamics was modeled comprising three states of Attitude errors and three states of Gyro biases. Kalman Filter was used to estimate the modeled error states. In results and discussion chapter 3, detailed analysis of Kalman filter based AHRS algorithm has been done in static and accelerated environments; it has been observed that Kalman filter based AHRS algorithm can filter out the noise both in gyroscopes and accelerometers and can estimate the gyroscope biases and attitude angles with much better accuracy. However, different dynamic conditions affect the gyro bias estimation. But, changing the tuning parameters, adaptively, can reduce this effect. It has been observed that to keep the attitude within certain accuracy, there is a certain time limit for which the accelerated environment can be tolerated.

8. References

[1] G. M. Siouris, *Aerospace avionics systems: a modern synthesis* vol. 466: Academic press New York, 1993.

[2] J. Farrell, *Aided navigation: GPS with high rate sensors*: McGraw-Hill New York, NY, USA:, 2008.

[3] W. Geiger, J. Bartholomeyczik, U. Breng, W. Gutmann, M. Hafen, E. Handrich, *et al.*, "Mems imu for ahrs applications," in *Position, Location and Navigation Symposium, 2008 IEEE/ION*, 2008, pp. 225-231.

[4] M. D. Shuster, "A survey of attitude representations," *Navigation*, vol. 8, p. 9, 1993.

[5] J. M. Cooke, "NPSNET: Flight simulation dynamic modeling using quaternions," Monterey, California. Naval Postgraduate School, 1992.
[6] J. Stuelpnagel, "On the parametrization of the three-dimensional rotation group," *SIAM review*, vol. 6, pp. 422-430, 1964.
[7] S. Emura and S. Tachi, "Sensor fusion based measurement of human head motion," in *Robot and Human Communication, 1994. RO-MAN'94 Nagoya, Proceedings., 3rd IEEE International Workshop on*, 1994, pp. 124-129.
[8] E. Foxlin, "Inertial head-tracker sensor fusion by a complementary separate-bias Kalman filter," in *Virtual Reality Annual International Symposium, 1996., Proceedings of the IEEE 1996*, 1996, pp. 185-194, 267.
[9] P. Setoodeh, A. Khayatian, and E. Frajah, "Attitude estimation by separate-bias Kalman filter-based data fusion," *Journal of Navigation*, vol. 57, pp. 261-273, 2004.
[10] F. L. Markley, J. L. Crassidis, and Y. Cheng, "Nonlinear attitude filtering methods," in *AIAA Guidance, Navigation, and Control Conference*, 2005.
[11] N. Wen-shu, W. Liao-ni, and L. Qi, "AHRS base on MEMS-IMU for aircraft model in wire-driven parallel suspension system," in *Mechanic Automation and Control Engineering (MACE), 2010 International Conference on*, 2010, pp. 3582-3585.
[12] H. Qi and J. B. Moore, "Direct Kalman filtering approach for GPS/INS integration," *Aerospace and Electronic Systems, IEEE Transactions on*, vol. 38, pp. 687-693, 2002.
[13] S. Han and J. Wang, "A novel method to integrate IMU and magnetometers in attitude and heading reference systems," *Journal of Navigation*, vol. 64, pp. 727-738, 2011.
[14] A. M. A. K. Marzouk, "A model based stroke rehabilitation monitoring system using inertial navigation systems," School of Engineering Science-Simon Fraser University, 2009.
[15] E.-H. Shin, "Estimation techniques for low-cost inertial navigation," *UCGE report*, 2005.
[16] M. S. Grewal and A. P. Andrews, "Kalman filtering: theory and practice using MATLAB, 2001," ed: John Wiley, New York.
[17] D. Titterton and J. Weston, "Strapdown Inertial Navigation Technology. 2-nd Edition," *The Institution of Electronical Engineers, Reston USA*, 2004.
[18] O. J. Woodman, "An introduction to inertial navigation," *University of Cambridge, Computer Laboratory, Tech. Rep. UCAMCL-TR-696*, 2007.
[19] R. Kleinbauer, "Kalman filtering implementation with Matlab," 2005.
[20] G. Welch and G. Bishop, "Course 8–An introduction to the Kalman filter," *SIGGRAPH 2001 Courses*, 2001.
[21] W. Li and J. Wang, "Effective Adaptive Kalman Filter for MEMS-IMU/Magnetometers Integrated Attitude and Heading Reference Systems," *Journal of Navigation*, vol. 1, pp. 1-15, 2012.